

SUBMISSION REQUIREMENT 1

Technical Approach

Proposed architecture and implementation approach for a secure South African application portal with staged review workflows, controlled document handling, and POPIA-aligned safeguards.

Project	Application Portal
Hosting Region	Azure South Africa North
Date	4 May 2026

1. Solution Overview

The system is structured around two main experiences: an applicant portal for registration, eligibility, consent, document upload, and submission; and a staff dashboard for review, scoring, shortlisting, interviews, and final decisions. A single application API sits at the centre and enforces permissions, workflow rules, calculations, and audit logging.

ASP.NET Core .NET 8

PostgreSQL

Azure App Service

Azure Blob Storage

Azure Key Vault

Azure Monitor

7

workflow stages

5+

staff roles supported

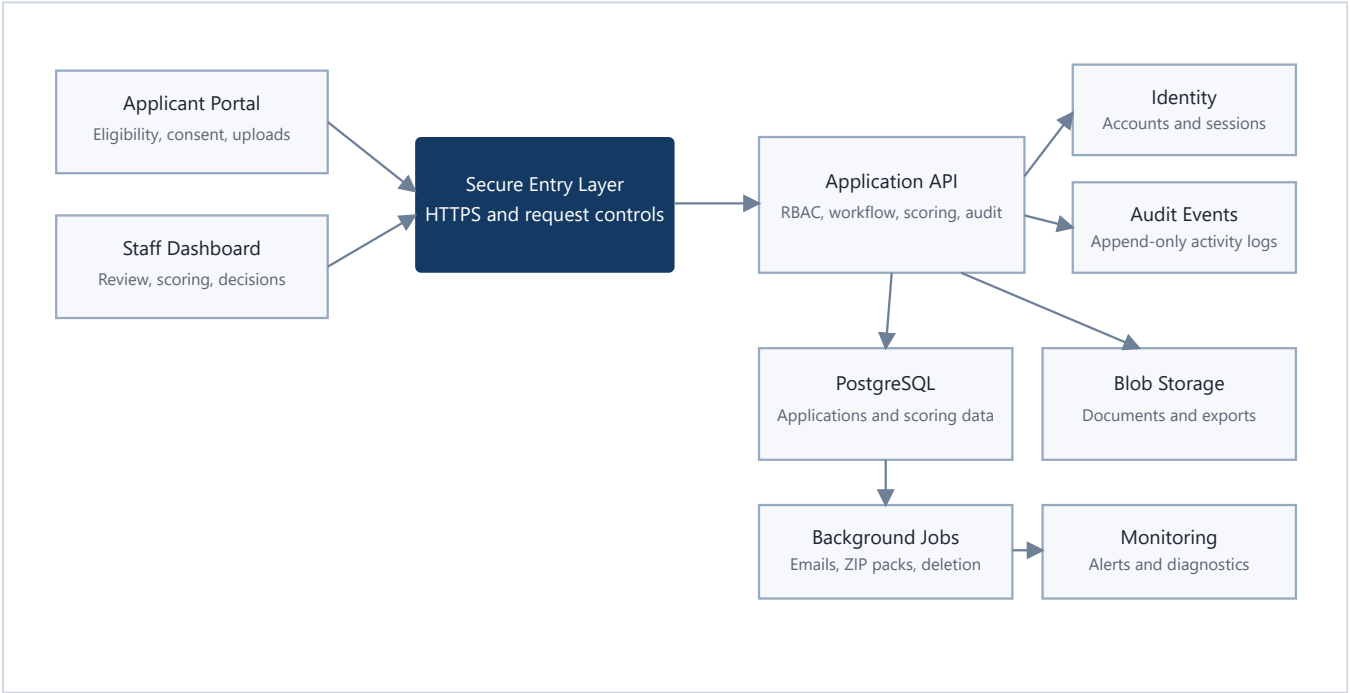
1

central workflow engine

100%

encrypted traffic and storage

2. High-Level Architecture



3. Role Access and Workflow Control

3.1 Role-Based Access

Role	Main Access	Restricted From
Essay Reviewer	Essay responses, applicant summary, own score entry	Academic marks, peer scores, shortlist and final outcomes
Academic Reviewer	Transcript capture, subject marks, calculated averages	Peer essay scores unless separately authorized
Document Checker	Required documents, checklist, issue notes	Scoring actions outside document verification
Screener	Combined scores, ranking, shortlist decision	Editing earlier reviewer-owned records
Interview Admin	Interview forms, score sheets, notes	Earlier stage editing controls
Final Decision Maker	Final review summary and final decision controls	Audit history modification

3.2 Workflow Enforcement

The application should move through explicit states. Shortlisting is blocked until essay review, academic scoring, and document verification are complete. Interview scoring is blocked unless the applicant is shortlisted. Final decisions can only be confirmed once the required earlier stages have been completed.

State	Control
Eligibility Failed	Reference number is kept for reporting, but the application does not proceed.
Submitted	Consent and all mandatory uploads must be present.
Essay Reviewed	Score stored with reviewer and timestamp.
Academics Scored	Subject marks and average preserved for later ranking.
Documents Verified	Checklist and document issues recorded.
Shortlisted	Allowed only when prior review stages are complete.

Interview Scored	Allowed only for shortlisted applicants.
Final Decision	Triggers outcome handling and communications.

3.3 Data Integrity

Submission Controls

- One active application per applicant per cycle
- Mandatory document validation before final submission
- Duplicate submission prevention

Reviewer Controls

- Concurrency checks on stage records
- Unique scoring records per reviewer and stage
- Refresh required if two staff actions conflict

4. POPIA-Aligned Controls

4.1 Data Protection Measures

Consent • Explicit consent before submission • Consent version and timestamp stored • Purpose tied to the application cycle	Encryption • TLS for all web traffic • Encrypted database and document storage • Keys and secrets managed securely
Audit Trail • User, role, action, target, and time captured • Downloads and exports logged • Administrative changes recorded separately	Data Subject Rights • Export of applicant data on request • Deletion or anonymisation after retention rules apply • Breach response support in the operating model

4.2 Core Data Structure

The database should separate each review stage into its own records so permissions remain clear and the history of each action is preserved cleanly.

users	roles	applications	consents	documents	essay_reviews	academic_scores
document_verifications	shortlist_decisions	interviews	final_decisions	audit_logs		

Workflow and permission enforcement sit primarily in the application layer, while the database supports transactional consistency and audit retention.